

COG-BCI Dataset

Overview

The **COG-BCI database** ([1]) is a multi-session, multi-task EEG dataset designed for research on **passive brain-computer interfaces** (pBCI). It provides simultaneous EEG recordings together with behavioral data across several well-established cognitive tasks, making it suited for studying cognitive workload, sustained attention, and vigilance in ecologically valid experimental settings.

In our local copy, the dataset is organized per subject and session, following a BIDS-like layout:

```
datasets/COG-BCI/
└─ sub-01/
   └─ ses-S1/
      ├── behavioral/  # trial-level behavioral logs (.mat)
      ├── chanlocs/    # channel locations (text)
      └─ eeg/          # raw EEG (.set / .fdt)
   └─ ses-S2/
   └─ ses-S3/
```

Each session contains the **same set of tasks**, with one EEGLAB pair (.set / .fdt) per condition.

Task battery

Task	EEG files	Measured construct
N-back (0/1/2-back)	zeroBACK oneBACK twoBACK	Working-memory load at three difficulty levels
Flanker	Flanker	Inhibitory control / response conflict
MATB	MATBeasy MATBmed MATBdiff	Multi-attribute task battery — sustained attention (easy / medium / difficult)
PVT	PVT	Psychomotor vigilance / fatigue
Resting state	RS_Beg_EC RS_Beg_E0 RS_End_Ec RS_End_E0	Eyes-closed / eyes-open baselines at session start and end

The behavioral logs mirror these conditions (0-Back.mat, 1-Back.mat, 2-Back.mat, Flanker.mat, MATB_Easy.mat, MATB_Med.mat, MATB_Diff.mat, PVT.mat), which enables joint analysis of neural and behavioral responses.

Recording setup

The montage (chanlocs/get_chanlocs.txt) corresponds to a **64-channel EEG cap** with scalp electrodes named after the 10-10 system (e.g. Fp1, Fz, F3, P0z, O1, O2), extended with:

- ECG1 — an ECG/EOG reference channel.
- nas, lhj, rhj — anatomical landmarks (nasion and left/right pre-auricular points) used for coordinate registration, not signal.

Channel coordinates are provided as 3D head-surface positions in millimeters, which is directly compatible with MNE's forward-modeling and plotting machinery.

Reading with MNE

Raw EEG is stored in EEGLAB format (.set + .fdt), readable with MNE via:

```
import mne
from nova2026.config import DATASET # points to a .set file

raw = mne.io.read_raw_eeglab(DATASET, preload=True)
print(raw.info) # channels, sampling rate, events
print(raw.annotations) # trial markers
```

See `src/nova2026/data/mne_reader.py` for the existing example in this repository.

Bibliography

References

- [1] M. F. Hinss, E. S. Jahanpour, B. Somon, L. Pluchon, F. Dehais, and R. N. Roy, *COG-BCI database: A multi-session and multi-task EEG cognitive dataset for passive brain-computer interfaces*. (Jul. 2022). Zenodo. doi: 10.5281/zenodo.6874129.